Module II — Theory of solids

Theory notes: concepts, results and what every symbol means

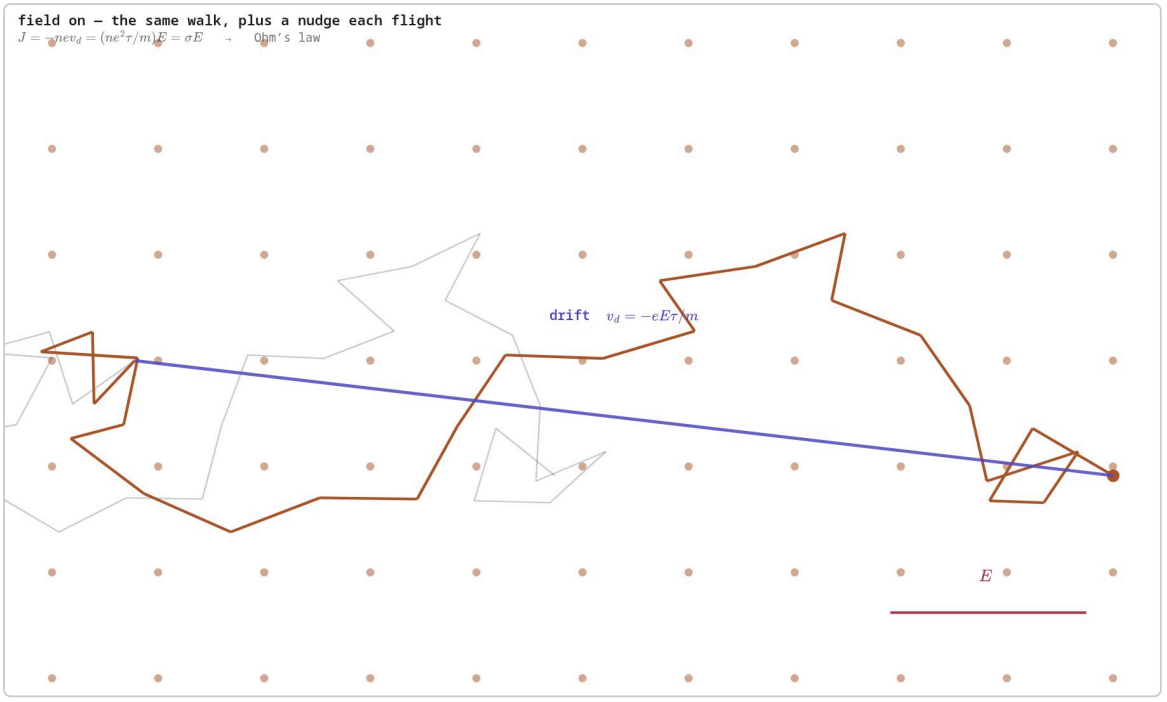
Free electron theory

IN ONE LINE Treat the metal's loose electrons as a gas of tiny balls. It explains conduction — but gets heat capacity wrong by a factor of 100, because it ignores the Pauli principle.

The Drude–Lorentz picture

A metal is a rigid array of positive ion cores with the outer (valence) electrons set loose to wander. The model assumes:

- Electrons move freely; the ion cores just sit there.
- Electrons ignore each other (no repulsion).
- The potential inside is **flat and constant**, taken as zero — so an electron's energy is purely kinetic. *This is the assumption that eventually breaks everything.*
- Outside is higher potential, so electrons stay trapped inside the metal.
- They behave like an ideal gas — but charged, and enormously denser than air ($n \sim 10^{28}$ per m^3).
- Lorentz treated that gas with ordinary **Maxwell–Boltzmann** statistics.



The whole classical model in one picture. Step 1 is the zero-field random walk — fast, and going nowhere. Step 2 adds the field: the walk is unchanged apart from a small nudge on each free flight, and those nudges accumulate into the slow **drift** that carries the current. The grey path is the same walk without the field, drawn for comparison — the difference between them is the entire current in a wire.

Three successes, four failures

What it gets right:

1. **Ohm's law.** With no field the motion is random and there is no current; apply a field and the electrons drift, scattering as they go, giving a steady current proportional to voltage.

2. **Wiedemann–Franz law.** The same electrons carry both charge and heat, so the ratio σ/K is the same constant for every metal at a given temperature.
3. **Optical properties.** Opacity and metallic shine — the electrons absorb the incoming wave, oscillate, and re-radiate it.

What it gets wrong:

1. **Resistivity vs temperature.** Predicts $\rho \propto \sqrt{T}$; experiment says $\rho \propto T$.
2. **Electronic heat capacity.** Equipartition gives every electron $\frac{3}{2}k_B T$, predicting $C_v = \frac{3}{2}R$ per mole. Measurements give **about 1% of that**, and it grows *linearly* with T.
3. **Magnetic susceptibility.** Predicts a strong $1/T$ (Curie) behaviour; the real thing is small and almost temperature-independent.
4. **Mean free path.** Cannot explain why, at low temperature, an electron travels past 10^8–10^9 atoms without scattering.

WHERE MARKS GET LOST

All four failures share one cause: **wrong statistics**. Maxwell–Boltzmann lets every electron soak up thermal energy; the Pauli exclusion principle does not. Sommerfeld fixes the first three by switching to Fermi–Dirac statistics. The fourth needs the *periodic lattice* — and that is band theory.

Fermi energy & friends

IN ONE LINE Trap electrons in a box, fill the levels two at a time from the bottom. The energy of the topmost filled level is the **Fermi energy** E_F, and it depends only on the electron density.

One dimension first

Put one electron on a line of length L with walls it cannot pass. Solving the Schrödinger equation gives the standing waves you would expect on a guitar string:

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\!\left(\frac{n\pi x}{L}\right), \qquad E_n = \frac{n^2 h^2}{8mL^2}$$

WHERE

$\psi_n(x)$	wavefunction of the nth state
E_n	energy of that state
n	quantum number: 1, 2, 3, ...
L	length of the box (wire)
x	position along the box
h	Planck's constant
m	electron mass

Now add N electrons. The **Pauli exclusion principle** allows two per level (spin up and spin down), so they stack up from the bottom and fill levels up to $n_F = N/2$. The energy of that top level is the Fermi energy:

$$E_F = \frac{h^2}{8m}\left(\frac{N}{2L}\right)^2$$

WHERE

E_F	Fermi energy: energy of the highest filled level at 0 K
N	number of electrons in the wire (two per level, so $N/2$ levels are filled)
L	length of the wire
h	Planck's constant
m	electron mass

Adding all the filled levels (using $\sum n^2 \approx s^3/3$) gives the average energy per electron in 1-D as $\bar{E}_0 = \frac{1}{3}E_F$. The 1-D density of states — the number of levels per unit energy — works out to $D(E) \propto E^{-1/2}$.

WHERE MARKS GET LOST

Watch the dimension. In 1-D $D(E) \propto E^{-1/2}$ (falling); in 3-D $D(E) \propto E^{+1/2}$ (rising). Same particle — only the geometry of counting states changes. Never quote one for the other.

Three dimensions and the Fermi sphere

In a cube of volume $V = L^3$ we use travelling waves and *periodic* boundary conditions (the crystal wraps around on itself — mathematically neater than hard walls):

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}}, \qquad k_x = \frac{2\pi n_x}{L} \text{ (and similarly for } y, z)$$

WHERE

$\psi_{\mathbf{k}}$	free-electron plane wave
V	volume of the box
$\mathbf{k}$	wave vector
$\mathbf{r}$	position
k_x	allowed value of the x component of $\mathbf{k}$
n_x	whole number $0, \pm 1, \pm 2, \ldots$
L	edge of the cubic box

So the allowed $\mathbf{k}$ values form a fine 3-D grid, each state occupying a k-space volume $(2\pi/L)^3$. The energy is $E = \hbar^2 k^2/2m$, so *states with the same energy lie on a sphere*. At $T = 0$ the electrons fill that sphere from the middle outwards, up to a radius k_F — the **Fermi sphere**.

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Typical metal values, worth knowing as sanity checks: $E_F \approx 2$–7 eV, $v_F \approx 10^6$ m/s, $T_F \approx 10^4$–10^5 K.

That last one is the key to Module II: T_F is *enormous* compared with room temperature, so 300 K is effectively "ice cold" for an electron gas — almost nothing is thermally excited.

The route for every numerical: structure → a → $n = (\text{atoms per cell} \times \text{valency})/a^3$ → k_F → E_F → v_F, T_F. Drill that chain; it is most of the marks in this module.

Density of states

IN ONE LINE $D(E)$ counts how many electron levels sit in each slice of energy. In 3-D it grows as $\sqrt{E}$, which makes the average electron energy $\frac{3}{5}E_F$.

What D(E) means and what it is

The **density of states** $D(E)$ is simply: *how many available levels lie between energy* E *and* $E + dE$. It matters because anything electrons do — absorb heat, conduct, get excited — depends on how many states are available to move into.

Differentiate the state count $N = \dfrac{V}{3\pi^2}\left(\dfrac{2mE_F}{\hbar^2}\right)^{3/2}$ with respect to energy:

$$D(E) = \frac{dN}{dE} = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2} \;\propto\; \sqrt{E}$$

WHERE

$D(E)$	density of states: number of electron states per unit energy, spin included
N	number of states below energy E
V	volume of the crystal
m	electron mass
$\hbar$	$h/2\pi$ (reduced Planck constant)
E	energy measured from the bottom of the band

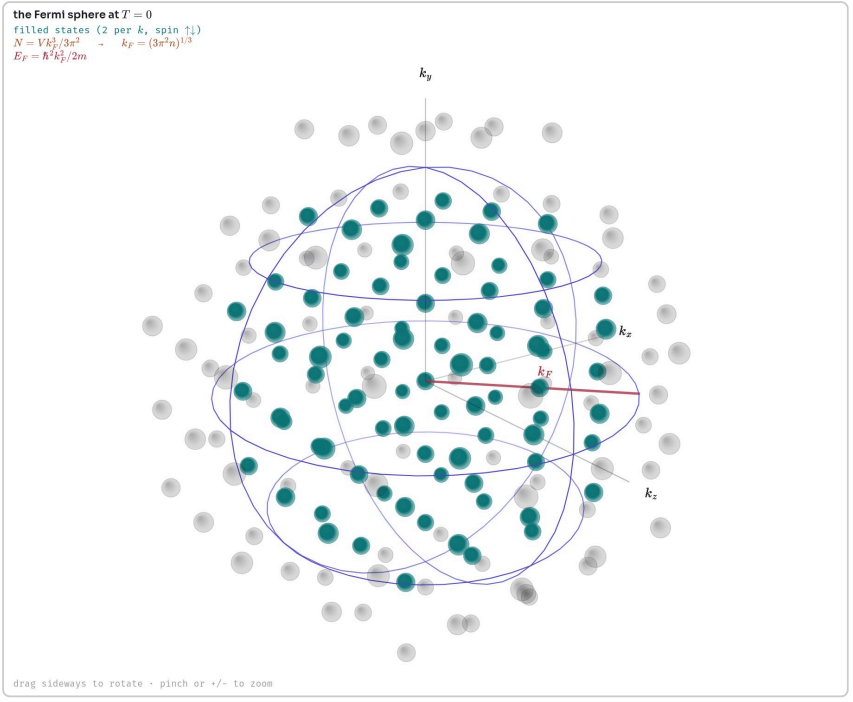
There is a neat shortcut for the value at the top. Since $N \propto E_F^{3/2}$, taking logs and differentiating gives:

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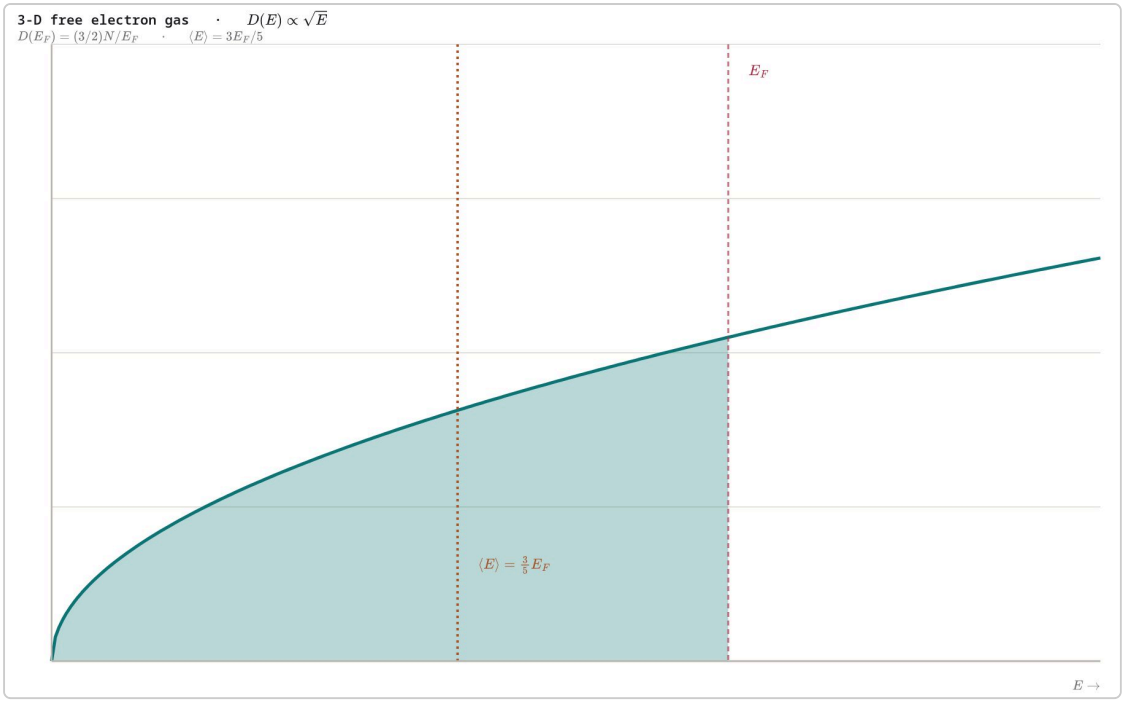
$$D(E_F) = \frac{3}{2}\frac{N}{E_F}$$

WHERE

$D(E_F)$	density of states exactly at the Fermi energy
N	total number of free electrons
E_F	Fermi energy



Spin it. The picture behind every Fermi-energy numerical. The allowed states form a grid in k-space; at $T = 0$ they fill from the middle outwards, and the surface where the filling stops is a sphere. Rotate it — the point of the free-electron model is that this region is a *ball*, so counting states is just a volume ÷ a volume.



Built in five steps — count states in k-space, change the variable to energy, differentiate, fill to E_F, then read off the two results. The chips switch the **dimensionality**, and that changes everything: in 3-D $D \propto \sqrt{E}$ is top-heavy and gives $\langle E \rangle = \frac{3}{5}E_F$; in 2-D D is flat and gives exactly $\frac{1}{2}E_F$; in 1-D $D \propto E^{-1/2}$ is bottom-heavy and gives $\frac{1}{3}E_F$. Those three numbers are three different exam questions.

Why the dimension matters — and the 1-D case they ask for

The shape of $D(E)$ is not a detail; it is decided by how much room a shell of k-space has at radius k, and that depends on the number of dimensions. Counting states out to k gives $N \propto k^d$ in d dimensions, and with $E \propto k^2$ that becomes $N \propto E^{d/2}$, so

$$D(E) = \frac{dN}{dE}; \propto; E^{\frac{d}{2}-1}$$

WHERE

$D(E)$	density of states
d	dimension of the system (1, 2 or 3)
E	energy

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1-D: $D(E) = \dfrac{L}{\pi}\left(\dfrac{2m}{\hbar^2}\right)^{1/2} E^{-1/2}$ — *falls* with energy, diverges at the band bottom.

2-D: $D(E) = \dfrac{Am}{\pi\hbar^2}$ — a constant, independent of E.

3-D: $D(E) = \dfrac{V}{2\pi^2}\left(\dfrac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$ — *rises* as $\sqrt{E}$.

And because the average energy is D-weighted, each shape drags the mean a different way: $\frac{1}{3}E_F$ in 1-D, $\frac{1}{2}E_F$ in 2-D, $\frac{3}{5}E_F$ in 3-D. If a question says “one-dimensional”, it is almost always asking for the $E^{-1/2}$ derivation or the $\frac{1}{3}E_F$ that follows from it.

Average energy = (3/5)E_F, and degeneracy pressure

Averaging the energy, weighted by how many states each energy has:

$$\bar{E}_0 = \frac{1}{N}\int_0^{E_F} E\,D(E)\,dE = \frac{\frac{2}{5}E_F^{5/2}}{\frac{2}{3}E_F^{3/2}} = \boxed{\frac{3}{5}E_F}$$

WHERE

$\bar{E}_0$	average energy per electron at 0 K
N	number of electrons
$D(E)$	density of states
E_F	Fermi energy (upper limit: states above it are empty at 0 K)

So even at absolute zero the electron gas is buzzing with kinetic energy — it cannot all settle to zero because Pauli forbids stacking everyone in the ground state. That leftover energy pushes outwards as a **degeneracy pressure**:

$$P = \frac{2}{5}\frac{NE_F}{V} = \frac{2}{3}\frac{E}{V}, \qquad B = \frac{5}{3}P = \frac{2}{3}nE_F$$

WHERE

P	pressure of the electron gas (it exists even at 0 K)
N	number of electrons
E_F	Fermi energy
V	volume
E	total energy of the electrons
B	bulk modulus: how hard the gas is to compress
n	electron density N/V

where B is the bulk modulus. These two lines are all you need for the compressibility numericals.

Fermi–Dirac & heat capacity

IN ONE LINE The Fermi function says a level is occupied with probability $f(E)$. Warming a metal only disturbs a thin sliver of width k_BT around E_F — which is why C_v is tiny and grows linearly with T.

The Fermi function

$$f(E) = \frac{1}{\exp\left(\dfrac{E - E_F}{k_BT}\right) + 1}$$

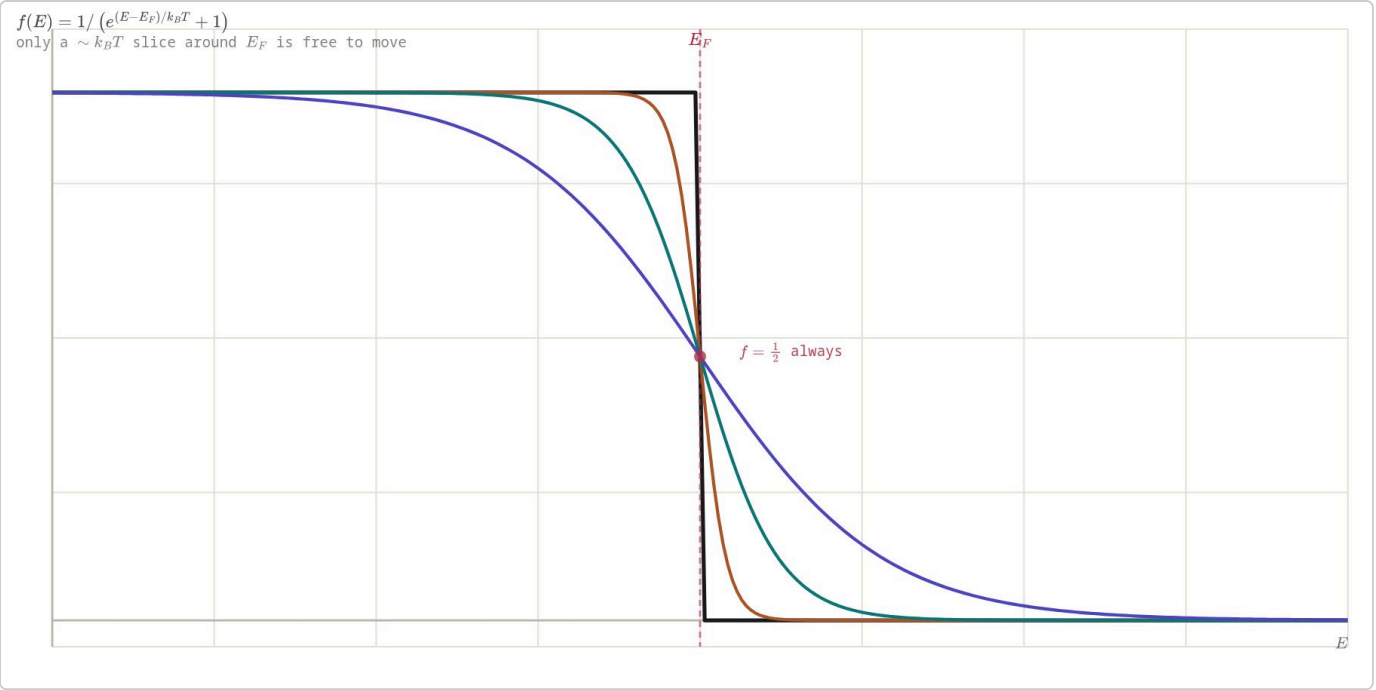
WHERE

$f(E)$	Fermi–Dirac function: probability that a state of energy E is occupied
E	energy of the state
E_F	Fermi level, where $f = \frac{1}{2}$
k_B	Boltzmann's constant
T	absolute temperature

Read it as: *the chance that a level of energy E actually has an electron in it*. That $+1$ in the denominator is the whole difference from classical statistics — it is what stops the probability ever exceeding 1, i.e. it is Pauli exclusion written as a formula.

- At $T = 0$ it is a perfect step: $f = 1$ below E_F, $f = 0$ above. Everything is filled to the brim and nothing above.

- At $T > 0$ the step softens over a width of about $k_B T$, and — worth memorising — $f(E_F) = \frac{1}{2}$ **at every temperature**.
- E_F itself barely moves with temperature: $E_F(T) \approx E_{F0}\left[1 - \frac{\pi^2}{12}\left(\frac{k_B T}{E_{F0}}\right)^2\right]$.



The step, and how little it softens. Four temperatures are plotted from the exact formula. Even at high T only a sliver of width $\sim k_B T$ around E_F is disturbed, and all four curves pass through $f = \frac{1}{2}$ exactly at E_F. Everything deep down stays frozen at $f = 1$ because Pauli gives it nowhere to go.

The Fermi level moves (slightly) with temperature. Above 0 K, E_F is no longer the top filled level but the level that is occupied with probability $\frac{1}{2}$. It falls a little as T rises:

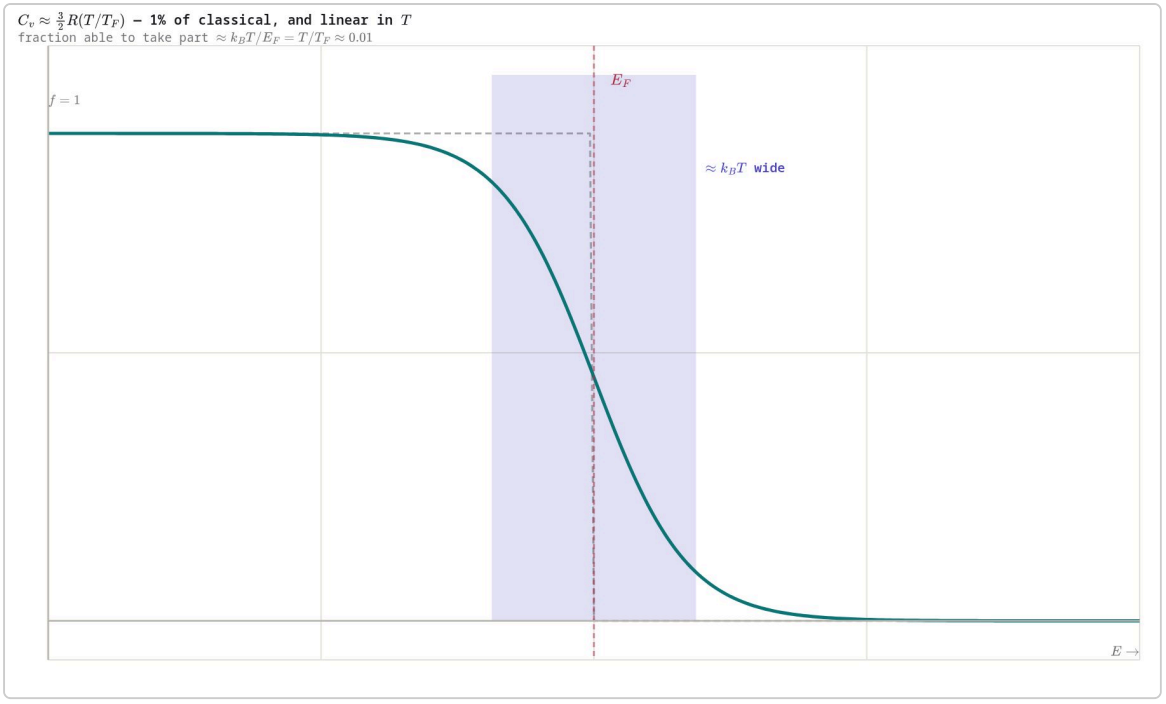
$$E_F \simeq E_{F0}\left[1 - \frac{\pi^2}{12}\left(\frac{k_B T}{E_{F0}}\right)^2\right]$$

WHERE

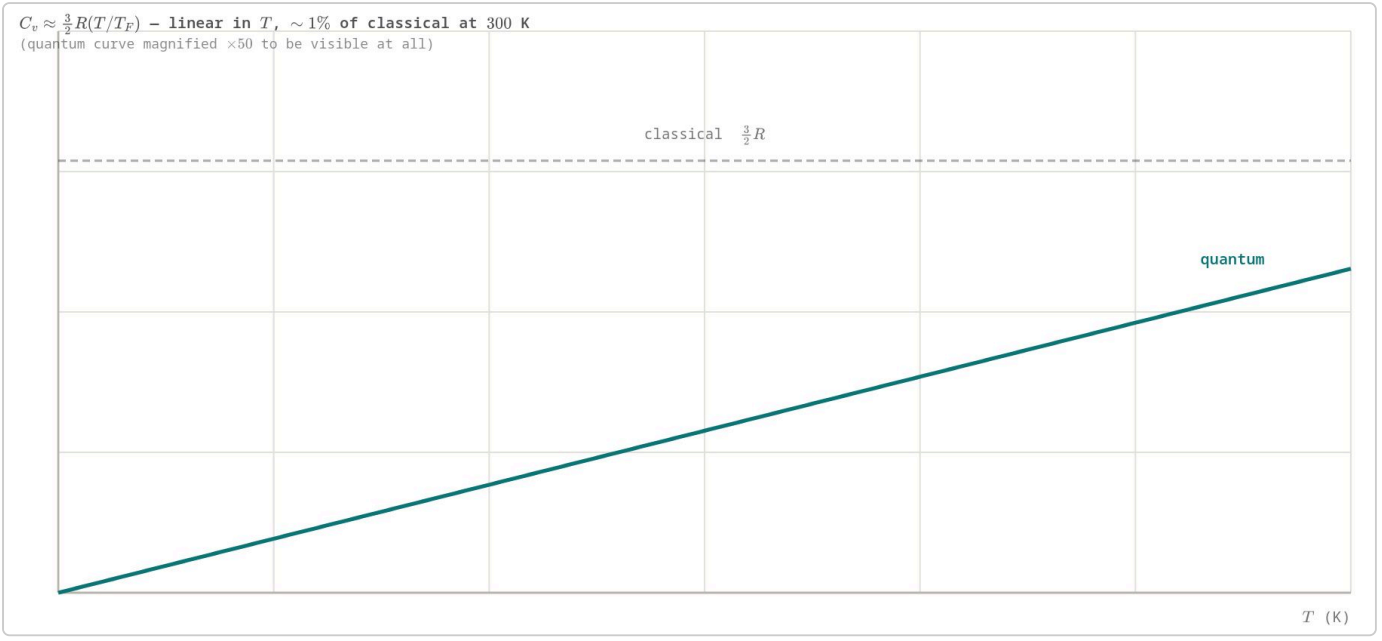
E_F	Fermi energy at temperature T
E_{F0}	Fermi energy at 0 K
$k_B T$	thermal energy; it is tiny next to E_{F0}, so the shift is very small

Because $k_B T \ll E_{F0}$ in a metal, the correction is tiny and E_F is usually taken as E_{F0}. Only electrons within about $k_B T$ of E_F (a fraction $k_B T / E_F \approx 0.01$ at room temperature for $E_F = 3$ eV) can be excited.

Why the electronic heat capacity is so small



The answer to “why is the electronic heat capacity so small?” in four steps. Everything hangs on the width of the softened region: it is about k_BT, and k_BT at room temperature is a fiftieth of an electron-volt against a Fermi energy of several. Only that sliver of electrons has anywhere to go.



The same result as a picture. The dashed line is what equipartition predicts — a constant $\frac{3}{2}R$ at every temperature. The teal curve is the quantum answer, **magnified 50×** just to be visible on the same axis: it is a straight line through the origin, so it vanishes as $T \to 0$ and never gets anywhere near the classical value at ordinary temperatures. Measure C_v at low T and you get $\gamma T + AT^3$ — the electron term linear, the lattice term cubic.

Both failures of the classical model are cured at once: the size is right, and the linear temperature dependence is right.

Band gaps & Bloch

IN ONE LINE Put the electron in a **repeating** potential instead of a flat one. Bragg reflection at the zone boundary opens gaps, and Bloch's theorem says the wavefunction is a plane wave times a lattice-periodic ripple.

Where the gaps come from

The free-electron model cannot explain three things: why solids split into conductors, semiconductors and insulators; why the Hall coefficient is sometimes *positive*; and why semiconductor resistivity falls as it warms.

The culprit is the flat-potential assumption. A real electron travels through a *repeating* row of positive ion cores, so the potential repeats too:

$$V(x) = V(x + a)$$

WHERE

- $V(x)$ potential energy of an electron at position x
- a lattice period; the potential repeats after every a

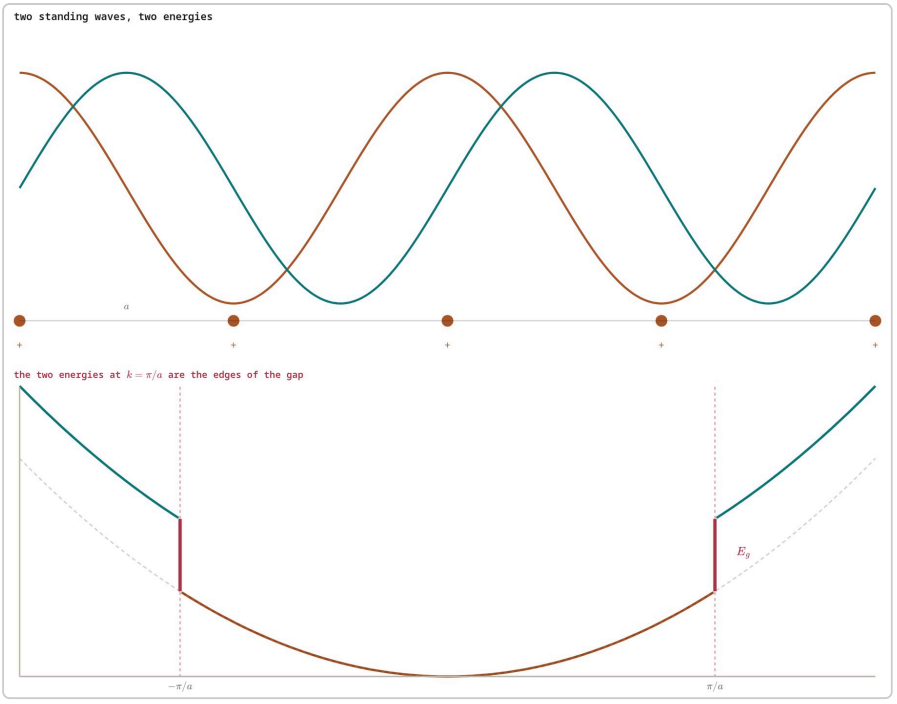
Once it repeats, an electron wave can Bragg-reflect off it — exactly as an X-ray does. That happens when

$$k = \frac{n\pi}{a} \qquad \text{(the Brillouin zone boundaries)}$$

WHERE

- k wave number at which Bragg reflection happens and an energy gap opens
- n non-zero whole number
- a lattice period

At those k the forward and reflected waves are equally strong and combine into **standing waves**. Two standing waves are possible: one piles the electron charge *on top of* the positive cores (lower energy, more attraction), the other piles it *between* them (higher energy). No state exists in between — and that missing range of energies is the **band gap**.



The most important picture in Module II. Step 2 is the Bragg reflection; steps 3 and 4 are the only two standing waves you can build from it, and the whole gap is the difference between where those two put their charge — on the cores, or between them. Step 5 shows the consequence on the $E(k)$ curve: two energies at one k, and nothing allowed in between.

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Gaps open at $k = \pm n\pi/a$, the Brillouin zone boundaries, because that is where Bragg reflection turns travelling waves into standing waves.

Bloch's theorem

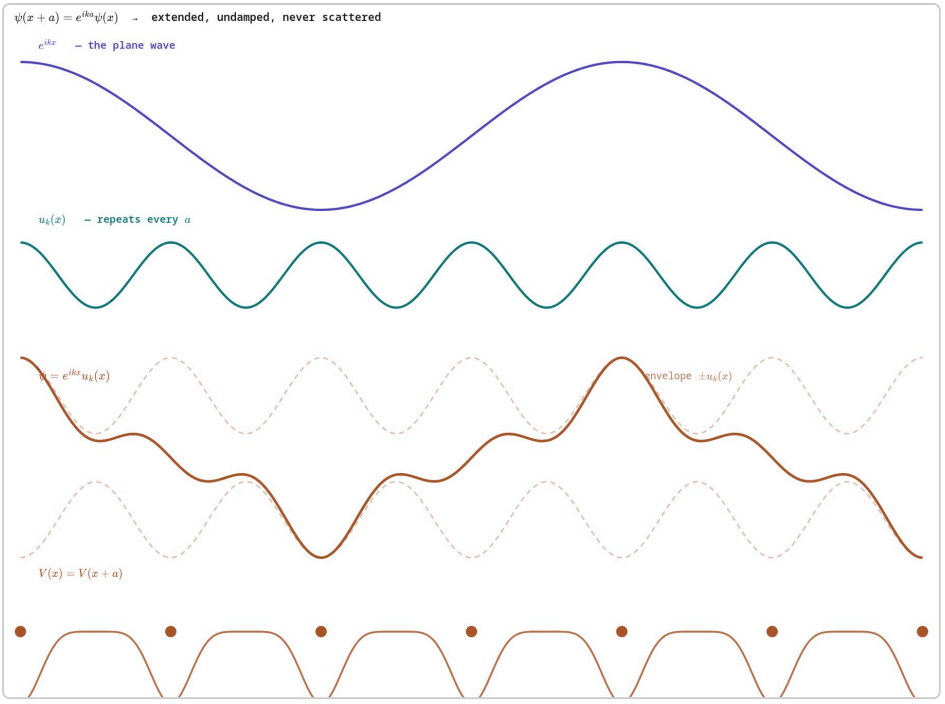
The theorem tells you the *shape* any wavefunction must have in a repeating potential:

$$\psi(x) = e^{ikx}u_k(x), \qquad \text{where } u_k(x) = u_k(x+a)$$

WHERE

$\psi(x)$	Bloch wavefunction
e^{ikx}	plane wave
k	wave number
$u_k(x)$	factor with the same period a as the lattice
a	lattice period

In words: a plane wave (the free-electron part) **multiplied by** a ripple that repeats with the lattice. The electron still travels through the crystal; it just gets modulated as it passes each atom.



The theorem as an assembly job: plane wave × lattice-periodic ripple = Bloch state. The dashed envelope at the last step is $\pm u_k(x)$ — note it repeats every a while the wave inside it does not, which is exactly what "$\psi(x+a) = e^{ika}\psi(x)$" means. Nothing about it decays with distance, and that is the sentence that answers the mean-free-path failure.

How the proof runs (enough to reproduce the argument):

1. The equation is second order, so it has two independent solutions f and g. Because V repeats, the shifted functions $f(x+a)$ and $g(x+a)$ must also be solutions, so they are combinations of f and g.
2. Requiring that shifting by a only multiplies ψ by a constant, $\psi(x+a) = \lambda\psi(x)$, gives a 2×2 determinant condition — a quadratic in λ.
3. The **Wronskian** $W = fg' - gf'$ is the same at x and $x+a$, which forces the constant term of that quadratic to be 1. So $\lambda_1\lambda_2 = 1$.

Two cases follow, and this is the punchline:

- **Roots complex,** $|\lambda| = 1$: then $\lambda = e^{\pm ika}$ with real k, giving $\psi(x+a) = e^{\pm ika}\psi(x)$ — exactly the Bloch form. The wave stays finite everywhere. These are the **allowed bands**.
- **Roots real,** $\lambda = e^{\pm\mu a}$: then ψ grows without limit at one end, so it cannot be normalised and is unphysical. These are the **forbidden gaps**.

Kronig–Penney

> IN ONE LINE A solvable repeating potential. One equation, whose left side must stay between -1 and $+1$ — inside that range you get an allowed band, outside it a gap.

The model and its one equation

Replace the real potential with a row of square wells and barriers of period $(a+b)$: zero inside the well of width a, height V_0 across the barrier of width b. Solve in each region, apply Bloch's theorem, match the wavefunction and its slope at the boundaries, and set the resulting determinant to zero.

In the useful limit where the barrier becomes very tall and very thin (with $V_0 b$ held fixed), the answer collapses to a single tidy condition:

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$$\boxed{P\,\frac{\sin(\alpha a)}{\alpha a}+\cos(\alpha a)=\cos(ka)} \qquad P=\frac{mV_0ba}{\hbar^2},\quad \alpha=\sqrt{\frac{2mE}{\hbar^2}}$$

WHERE

P	strength of the barriers $= mV_0ba/\hbar^2$
α	$\sqrt{2mE}/\hbar$: wave number of the electron between barriers
E	electron energy
a	width of the well between barriers
b	width of a barrier
V_0	height of a barrier
m	electron mass
k	Bloch wave number
$\hbar$	reduced Planck constant

P is the dimensionless *barrier strength* — how tightly the lattice grips the electron.

Reading bands and gaps off it

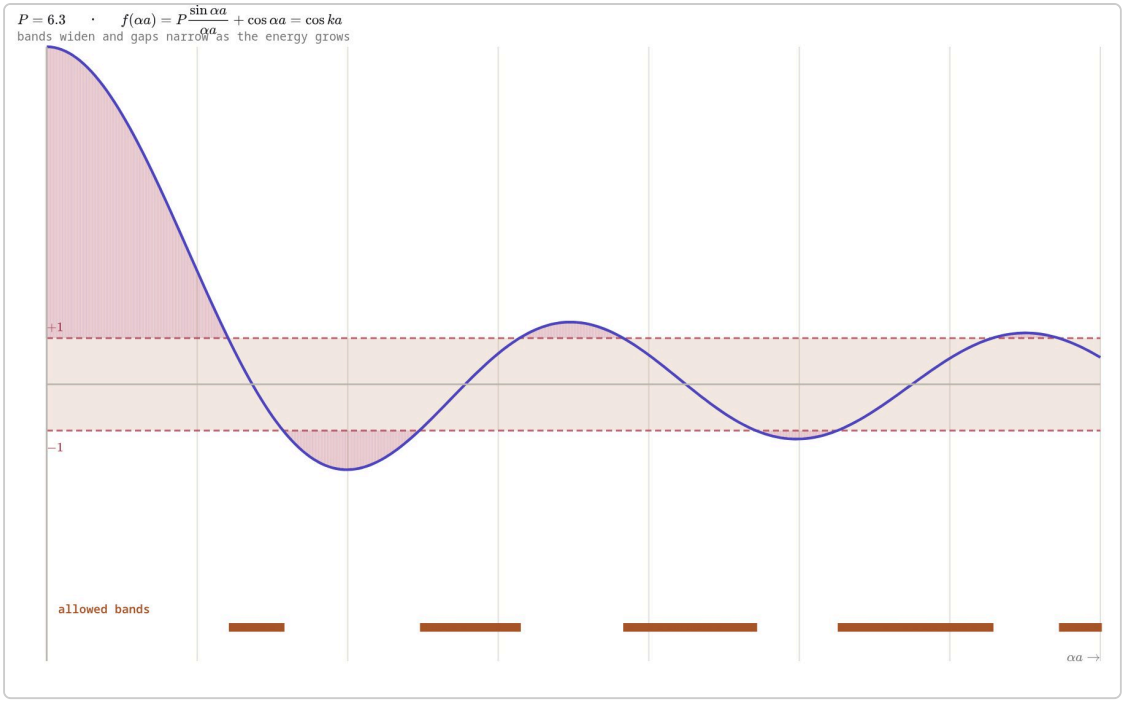
Here is the trick that makes the whole model work. The right-hand side is a cosine, so it **can never leave the range -1 to $+1$**. A real value of k therefore exists only when

$$-1 \le P\frac{\sin(\alpha a)}{\alpha a}+\cos(\alpha a) \le 1$$

WHERE

P	barrier strength
αa	$\sqrt{2mE}\,a/\hbar$, a dimensionless measure of the energy E
rule	$\cos ka$ cannot go beyond ± 1, so only energies where the left side stays within $[-1, 1]$ exist: allowed bands. The rest are gaps.

Energies satisfying that are **allowed**; energies where the left side escapes the range are **forbidden**. Plot the left side and the band structure simply appears.



The whole band structure in one curve, built in four steps. Then use the chips on P: at $P \to 0$ the curve never leaves the corridor and there are *no gaps at all* — the free electron is back. Crank P up and each allowed band is squeezed towards a single sharp level — isolated atoms. Those are the two limits the paper asks for, and here you can watch them happen.

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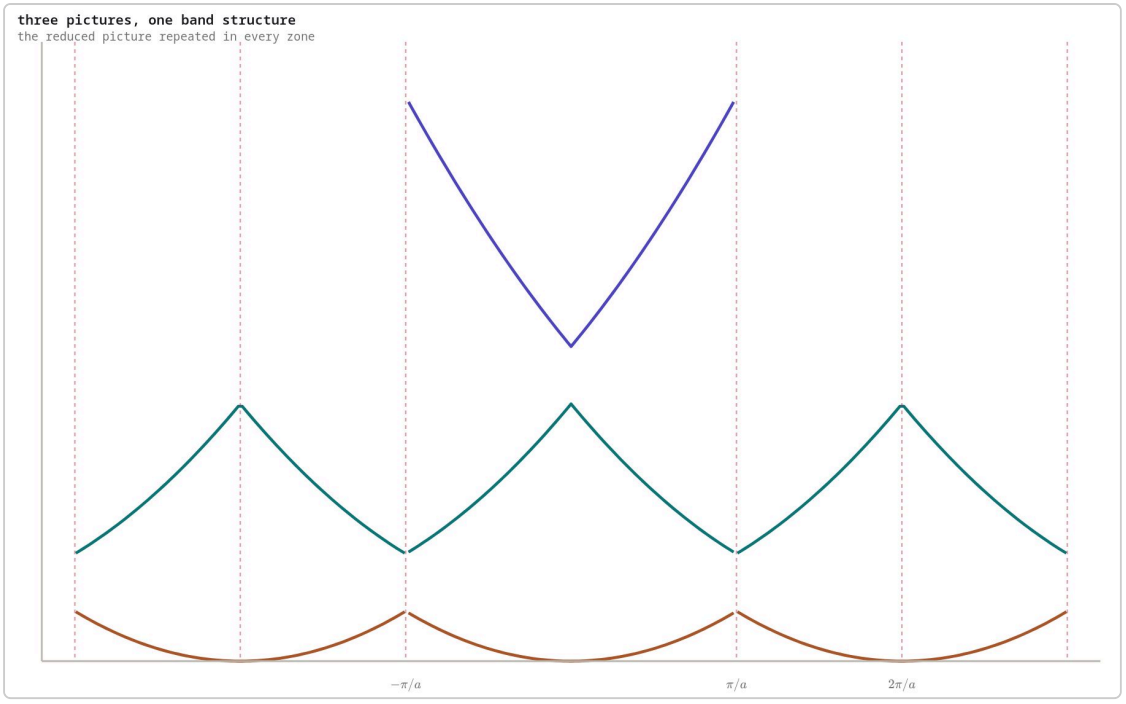
The two limits — very commonly asked:

- $P \to \infty$ (huge barriers, isolated atoms): needs $\sin(\alpha a) = 0$, so $\alpha a = n\pi$ and $E_n = \dfrac{n^2 h^2}{8ma^2}$. The bands shrink to sharp atomic levels.
- $P \to 0$ (no barriers, free electrons): gives $\cos(\alpha a) = \cos(ka)$, so $\alpha = k$ and $E = \dfrac{\hbar^2 k^2}{2m}$. The gaps vanish and the free-electron parabola is back.

Also: as energy rises, the $1/\alpha a$ factor shrinks the oscillation, so **bands get wider and gaps get narrower**.

How the bands are drawn, and how many electrons fit

Since $\cos(ka)$ repeats every $2\pi/a$, $E(k)$ is periodic in k-space. Three standard ways to draw it: the **extended** scheme (band n drawn in zone n), the **reduced** scheme (everything folded back into the first zone), and the **periodic** scheme (repeated in every zone). Same information, three pictures.



All three zone schemes, one per step — extended, reduced, periodic. They contain identical information; only the bookkeeping changes. Step through them once and the exam question “which scheme is this, and what does it show?” becomes free marks.

Now count the states. For a crystal of N cells and length $L = Na$, periodic boundary conditions give $k = 2\pi n/L$, so the number of k states in one zone (width $2\pi/a$) is $\dfrac{L}{2\pi} \cdot \dfrac{2\pi}{a} = \dfrac{L}{a} = N$. With two spins:

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One band holds exactly $2N$ electrons, where N = number of primitive cells. One valence electron per atom → band half full → metal. Two per atom → band exactly full → insulator, *unless* bands overlap.

Effective mass & classification

IN ONE LINE An electron in a band responds to a field as if it had a different mass $m^* = \hbar^2/(d^2E/dk^2)$ — which can even be negative. A full band carries no current at all.

Speed and effective mass

An electron in state k is really a wave packet, and it travels at the **group velocity**:

$$v = \frac{1}{\hbar}\frac{dE}{dk}$$

WHERE

v	group velocity of the electron (speed of the wave packet)
$E(k)$	band energy as a function of k
k	wave number
$\hbar$	reduced Planck constant
ω	angular frequency, $E/\hbar$

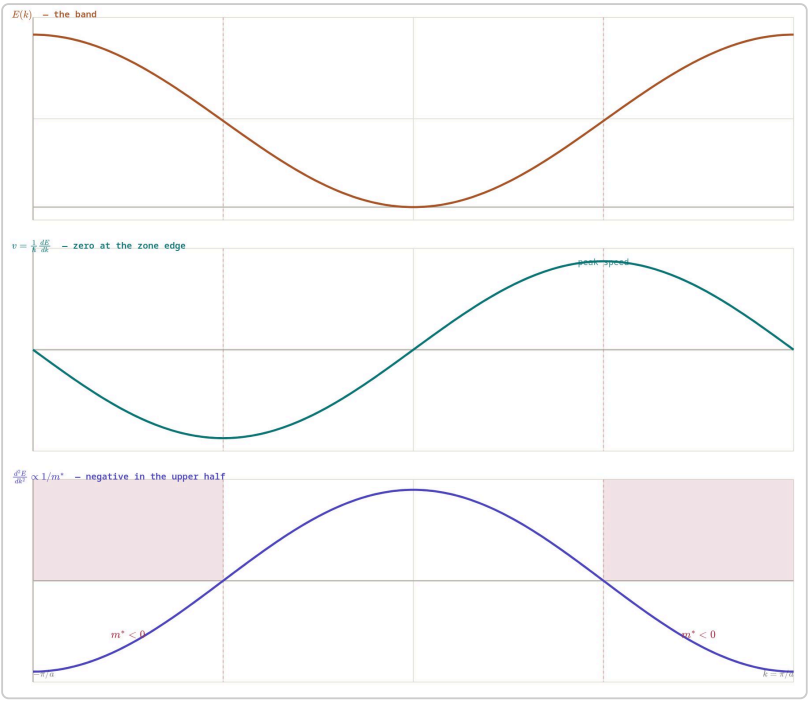
So the *slope* of the band gives the speed. At the bottom of a band ($k = 0$) and at the zone boundary ($k = \pm\pi/a$) the slope is zero, so $v = 0$ — those are the standing waves again. The speed peaks in between, at the point of steepest slope.

Reading m^* off a band — the practical version

You are never asked to compute m^* from first principles; you are asked to *read* it. Three rules cover every question:

- **Curvature is everything.** $m^* = \hbar^2 \Big/ \dfrac{d^2E}{dk^2}$, so a sharply curved band means a light, responsive electron and a flat band means a heavy, sluggish one. Narrow bands (weakly overlapping atomic orbitals) therefore give large effective masses — that is why d-band electrons in transition metals are heavy.
- **Bottom of a band: m^* small and positive.** Here $E \approx E_0 + \hbar^2 k^2 / 2m^*$ and the electron behaves exactly like a free particle that happens to have the wrong mass. This is the region that matters for a semiconductor's conduction band.
- **Top of a band: m^* negative.** The curvature has flipped, and the electron accelerates against the applied force. This is the region that matters for holes.

In between, at the inflection point $k = \pm\pi/2a$, the curvature passes through zero and m^* formally *diverges* — the electron there refuses to accelerate at all, because the lattice takes back exactly as much momentum as the field supplies.



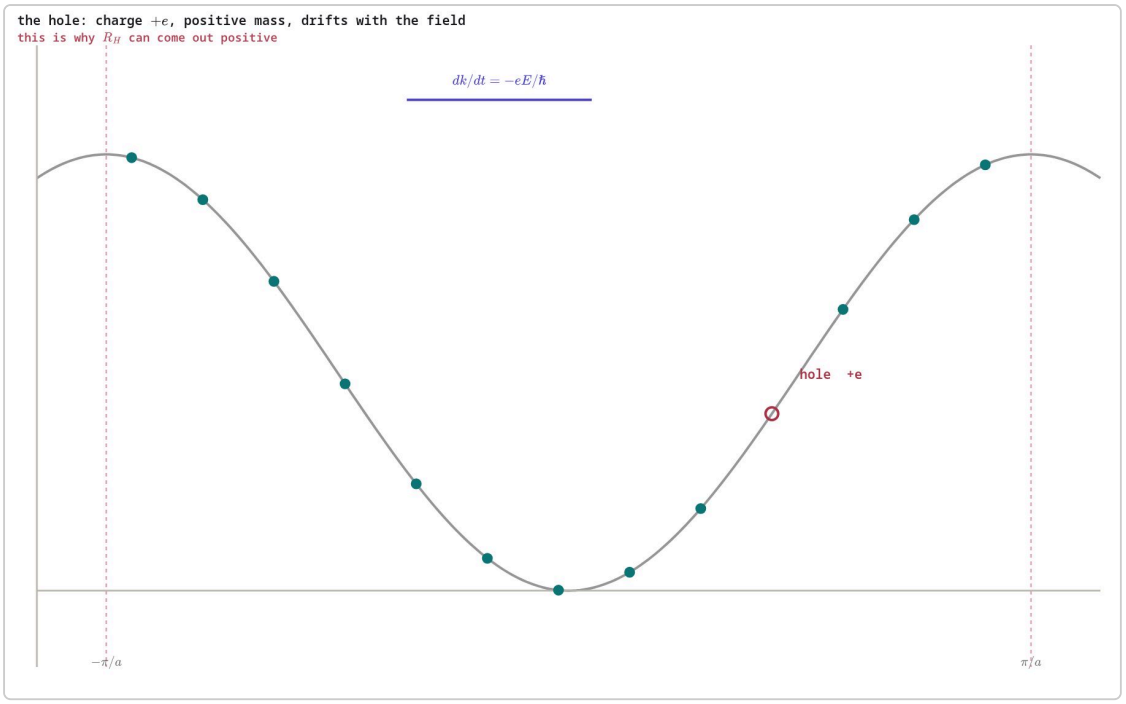
One band, differentiated twice — one panel per step. Top: $E(k)$. Middle: the slope, giving the velocity — zero at both ends, maximum at the inflection. Bottom: the curvature, which is proportional to $1/m^*$ — positive in the lower half of the band, zero at the inflection (so $m^* \to \infty$ there), and negative in the upper half.

- **Lower half of the band:** curvature positive → $m^* > 0$, the electron accelerates the way you push it.
- **At the inflection:** curvature zero → $m^* \to \infty$, the field cannot change its speed at all.
- **Upper half:** curvature negative → $m^* < 0$. The electron accelerates *backwards*.

WHERE MARKS GET LOST

Negative mass looks like nonsense but breaks nothing. m^* is a bookkeeping trick that hides the lattice force: the electron really feels both the applied field *and* the lattice. Near the top of a band the lattice force (Bragg reflection handing momentum back to the crystal) beats the applied force, so the net acceleration opposes the field — and we record that as $m^* < 0$. Total momentum, electron plus lattice, is conserved throughout.

Holes



Where the fictitious positive particle comes from. The key move is step 2: instead of adding up the currents of every remaining electron, subtract the one that is missing. Because that missing electron sat where $m^* < 0$, the two minus signs cancel and the vacancy behaves like a genuinely positive particle. This is the whole content of "holes", and it is why a Hall measurement can report the wrong sign of carrier.

A nearly full band with one electron missing is far easier to describe by tracking the *vacancy* instead of the 10^{23} remaining electrons. Because the missing electron sat where m^* is negative, the vacancy — a **hole** — behaves as a real particle with

$$m_h^* = -m_e^* > 0, \qquad \mathbf{k}_h = -\mathbf{k}_e, \qquad \mathbf{v}_h = \mathbf{v}_e, \qquad q_h = +e$$

WHERE

$m_e^*,\ m_h^*$	effective masses of the missing electron and of the hole
$\mathbf{k}_e,\ \mathbf{k}_h$	wave vectors of the missing electron and of the hole
$\mathbf{v}_e,\ \mathbf{v}_h$	their velocities
q_h	charge of the hole, $+e$

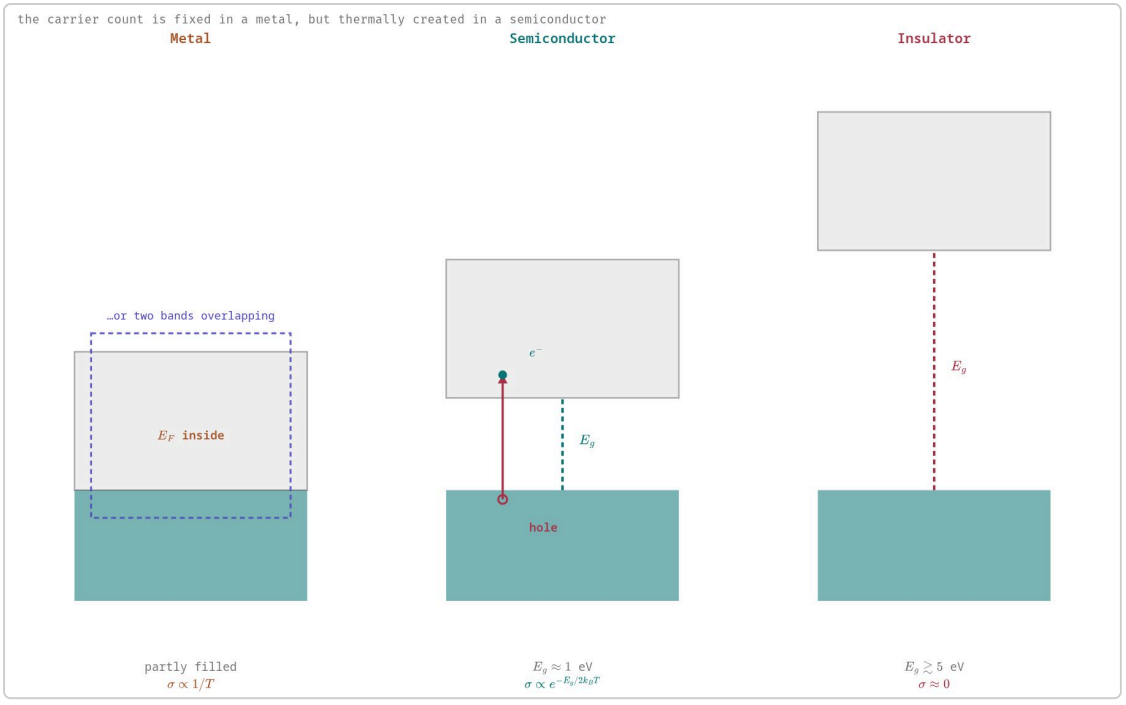
A positive charge with a positive mass. This is what makes the Hall coefficient come out positive in some materials — something free-electron theory could never explain.

Metals, semiconductors, insulators

First the key result: **a completely full band carries no current.** Summing the contributions of all its electrons gives a total proportional to dE/dk evaluated at the zone boundary — and that slope is zero. So no matter how hard you push, a full band contributes nothing. *Conduction requires a partly filled band.*

LEARN THIS BY HEART

The one-line test. A solid conducts if and only if there is an *empty* level immediately above an *occupied* one. Partly filled band, or two bands overlapping → metal. Full band with a big gap above it → insulator. Full band with a small gap above it → semiconductor, which is an insulator at $T = 0$ and a conductor when warmed.



The three cases. Teal = filled valence band, outlined = empty conduction band. A metal has no gap at the top of the filled states, so empty levels sit immediately above occupied ones. A semiconductor has a narrow gap that thermal energy can bridge. An insulator's gap is far beyond $k_BT \approx 0.026$ eV at room temperature.

1. **Insulator** — full valence band, empty conduction band, wide gap $E_g \gtrsim 5$–6 eV (diamond). Thermal energy cannot lift electrons across. Essentially no conduction.
2. **Semiconductor** — same picture but a narrow gap, $E_g \sim 1$ eV (Si 1.1, Ge 0.7 eV). At $T = 0$ it *is* an insulator. Warm it and electrons hop the gap leaving holes behind, with

$$\sigma \propto e^{-E_g/2k_BT}$$

WHERE

σ	conductivity of the intrinsic semiconductor
E_g	band gap
k_B	Boltzmann's constant
T	temperature

   — conductivity rises steeply with temperature.
3. **Metal** — a partly filled band (one electron per atom filling half of the $2N$ capacity), or two bands that overlap (the alkaline earths). Empty states sit right next to filled ones, so conduction is easy. Here $\sigma \propto 1/T$: warming adds lattice vibrations that scatter electrons more.